

PH.D. QUALIFYING EXAM 2020: PROBABILITY

Rules: This is a four hour exam consisting of two problems, both of which should be answered. You can take the exam at any location where you have the facility to scan or take pictures of your answers and convert them into pdf files. You are allowed to use any material of your choice prepared before the time of the exam, but use of online materials or consulting anyone else during the exam is not be permitted. If you have any questions during the exam, please email Professors Harry Crane (hcrane@stat.rutgers.edu) and Tirthankar Dasgupta (td370@stat.rutgers.edu).

Please make sure to **handwrite** your answers clearly and legibly on blank papers, preferably using pens, leaving an impression deep enough for the papers to be scanned. **Typed answers are not permitted.**

Create two pdf files for your answers - one for each problem. Both the pdf files containing your answers should be emailed to td370@stat.rutgers.edu and hcrane@stat.rutgers.edu by 2:15 PM. **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Good luck!

Problem 1

Note: This problem is about convergence properties of triangular arrays of random variables and their applications. A triangular array of random variables is a doubly indexed sequence of random variables of the form

$$\begin{array}{ccccccc} & & X_{1,1} & & & & \\ & & X_{2,1} & X_{2,2} & & & \\ & \cdots & \cdots & \cdots & \cdots & & \\ & X_{n,1} & X_{n,2} & X_{n,3} & \cdots & X_{n,n} & \\ & \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \end{array}$$

PART I: Convergence properties

Question 1 (WLLN for row-iid triangular arrays): For $n = 1, 2, \dots$ and $j = 1, \dots, n$, let $\{X_{n,j}\}$ be a triangular array of random variables such that for each n the random variables $X_{n,1}, \dots, X_{n,n}$ are assumed to be independent and identically distributed. Let $S_n = X_{n,1} + \dots + X_{n,n}$. Show that, if $\sup_n E(X_{n,j}^2) < \infty$, then $\frac{S_n - E(S_n)}{n} \xrightarrow{p} 0$.

Question 2 (Deriving iid-version of regular CLT from CLT for triangular arrays):

The triangular array CLT under Lindeberg's condition is stated below:

Theorem 1. For $n = 1, 2, \dots$ and $j = 1, \dots, n$, let $\{X_{n,j}\}$ be a triangular array of random variables such that for each n the random variable $X_{n,1}, \dots, X_{n,n}$ are assumed to be independent with $E(X_{n,j}) = 0$ and $\text{var}(X_{n,j}) = \sigma_{n,j}^2 < \infty$, $j = 1, \dots, n$. Let $S_n = X_{n,1} + \dots + X_{n,n}$, $s_n^2 = \text{var}(S_n)$ and $Z_n = S_n/s_n$. If for each $\epsilon \in (0, 1)$,

$$(1) \quad \text{Lind}_{\epsilon,n} \equiv \sum_{j=1}^n E \left[\left(\frac{X_{n,j}}{s_n} \right)^2 I \left(\frac{|X_{n,j}|}{s_n} > \epsilon \right) \right] \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

then $Z_n \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$.

Now recall the regular CLT for iid random variables, which states:

Theorem 2. Let $Y_1, Y_2, \dots$ be iid with $E(Y_j) = 0$ and $\text{var}(Y_j) = \sigma^2 < \infty$ for $j = 1, 2, \dots$. Let $\tilde{S}_n = \sum_{j=1}^n Y_j$, $\tilde{s}_n^2 = \text{var}(\tilde{S}_n) = n\sigma^2$ and $\tilde{Z}_n = \tilde{S}_n/\tilde{s}_n = \tilde{S}_n/\sqrt{n}\sigma$. Then $\tilde{Z}_n \xrightarrow{d} N(0, 1)$ as $n \rightarrow \infty$.

Show that Theorem 2 is a special case of Theorem 1. In other words, prove Theorem 2 assuming Theorem 1 is true.

Hint: Create a connection between $X_{n,j}$ in Theorem 1 and Y_j in Theorem 2.

PART II: An application of the triangular array CLT

Background: For each $N = 1, 2, \dots$, consider a population π_N of N individuals or units for which the values of a variable Y of interest are $Y_{N,1}, \dots, Y_{N,N}$. For example, the individuals can be human beings and Y can represent their weights or heights. Note that (a) $Y_{N,1}, \dots, Y_{N,N}$ are fixed quantities and not random variables and (b) two indices are used in the suffix of $Y_{N,j}$ to allow representation of a sequence of populations $\{\pi_N, N = 1, 2, \dots\}$. We are interested in making inference about the population total $\sum_{i=1}^N Y_{N,i}$ using its sample counterpart. A random sample of size n drawn from the population using Bernoulli sampling: For $N \geq 1$, each of the units in π_N is included in the sample independently with probability n/N , for some $1 \leq n \leq N$. That is, the j th unit of π_N is included in the sample if and only if $K_j = 1$ for $K_1, K_2, \dots, K_N \stackrel{iid}{\sim} \text{Bern}(n/N)$. Consequently, the sample counterpart or estimator of $\sum_{i=1}^N Y_{N,i}$ is $\sum_{i=1}^N K_i Y_{N,i}$, and in this problem we will explore the asymptotic distribution of this estimator.

Question 3:

- (a) Define $X_{N,j} = (Y_{N,j} - \bar{Y}_N)(K_j - n/N)$ for $j = 1, \dots, N$ where $K_1, \dots, K_N$ are defined above and $\bar{Y}_N = N^{-1} \sum_{j=1}^N Y_{N,j}$ is the population mean. Find the expectation and variance of $X_{N,j}$.
- (b) Are $X_{N,1}, \dots, X_{N,N}$ independent? Give a proof.
- (c) Show that the triangular array $\{X_{N,j}\}$ satisfies Lindeberg's condition (1) if

$$(2) \quad \frac{\max_{1 \leq j \leq N} (Y_{N,j} - \bar{Y}_N)^2}{\sum_{j=1}^N (Y_{N,j} - \bar{Y}_N)^2} \max \left(\frac{N-n}{n}, \frac{n}{N-n} \right) \rightarrow 0,$$

as both n and $N - n$ tend to infinity.

- (d) Derive the asymptotic distribution of $T_N \equiv \sum_{j=1}^N K_j (Y_{N,j} - \bar{Y}_N)$ as n and $N - n$ tend to infinity, assuming (2) is true.

Problem 2

This problem concerns a population for which each individual can at any time be in one of three states, A , B and C . The population evolves in discrete time according to the following rules:

- (1) Each individual x in state A in generation n produces a random number ξ_x of offspring in generation $n+1$, all of type B . Individual x then dies. The distribution of ξ_x is
- (3)
$$\mathbb{P}(\xi_x = k) = \alpha^k(1 - \alpha), \quad k = 0, 1, 2, \dots,$$
for $0 < \alpha < 1$.
- (2) Each individual in state B in generation n mutates to either state A or state C in generation $n+1$. The probability of mutation to A and C , respectively, are p and $q = 1 - p$ for some $0 < p < 1$.
- (3) Each individual in state C in generation n dies.
- (4) All individuals in generation n act independently of one another and of the past.

As stated above, the parameters satisfy $0 < \alpha < 1$ and $0 < p < 1$. For $i = A, B, C$, let Z_n^i be the number of individuals in state i in generation n . Assume that at time $n = 0$ there is a single individual in state A .

Question 4:

- (a) Show that with probability 1, either $Z_n^A = Z_n^B = Z_n^C = 0$ for all large n or $\lim_{n \rightarrow \infty} (Z_n^A + Z_n^B + Z_n^C) = \infty$.
- (b) Express $E(Z_n^A)$, $E(Z_n^B)$ and $E(Z_n^C)$ as functions of α , p and n .
- (c) Let

$$p_e = \mathbb{P}(Z_n^A = Z_n^B = Z_n^C = 0 \text{ for all large } n)$$

denote the probability of extinction. For which values of α and p is $p_e = 1$? Give a proof.

- (d) In the case $p_e < 1$, find an explicit closed form expression for p_e in terms of α and p .
- (e) Define sequences $\{a(n)\}_{n \geq 1}$ and $\{b(n)\}_{n \geq 1}$ such that

$$Y_n := \frac{Z_n^B}{b(n)}, \quad n \geq 1,$$

converges almost surely to a finite limiting random variable. Give a proof of convergence for $\{Y_n\}_{n \geq 1}$. (Note that for this part you are expected to find explicit forms of $\{a(n)\}_{n \geq 1}$ and $\{b(n)\}_{n \geq 1}$ that guarantee convergence of $\{Y_n\}_{n \geq 1}$ and justify your choice of $\{a(n)\}$ and $\{b(n)\}$ by proving convergence.)